OLEG A. STARYKH

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Education

Ph.D. in Physics, October 1991:
Institute for High Pressure Physics, Russian Academy of Sciences, Moscow.
M.S. in Physics and Engineering, June 1988: *Diploma with honors*,
Moscow Institute of Physics and Technology (MIPT), Moscow, Russia.

Honors and Awards

— ∗ Co-PI, with A. Keselman (Technion, Israel), United States-Israel Binational Science Foundation **BSF** award # 2024200, amount $\$90,000$, 10/01/2025-9/1/2027,
project title: *Quantum phases and excitations of magnetized frustrated magnets.*
— *Outstanding Referee* for the journals of the American Physical Society, **2022**.
— **Fellow** of the American Physical Society, **2020**.
— ∗ **NSF DMR**, award # 1928919, amount $\$330,000$, 12/01/2019-11/30/2023:
project title: *Collective Modes and Electrodynamics of Interacting Spin Liquids.*
— ∗ **Japan Society for the Promotion of Science** FY2017 Invitational award:
Short term visit – travel and daily expenses for 19 days (Nov 3-22, 2017).
— ∗ **NSF DMR**, award # 1507054, amount $\$306,000$, 3/2016-11/2019:
• project title: *Order-by-frustration: emergent condensed states of frustrated magnets.*
— ∗ **NSF DMR**, award # 1206774, amount $\$270,000$, 9/2012-8/2016:
• project title: *Frustrated magnetism in spin-orbit coupled materials.*
— ∗ Co-PI (together with Prof. Pesin) of Utah MRSEC Seed Grant, amount $\$24,000$, 2014:
• project title: *Transport and correlated phases in system with strong spin-orbit interaction*
— ∗ **NSF DMR**, award # 0808842, amount $\$210,000$, 5/2009-4/2013:
• project title: *Competing orders in frustrated magnets and nanostructures.*
— ∗ **The Petroleum Research Fund**, American Chemical Society, amount $\$80,000$, 9/2005-8/2007.
• project title: *Frustrated magnetism as a problem of coupled spin chains.*
— **ITP Scholar** for 2001-2003, KITP, University of California, Santa Barbara.
— ∗ **Cottrell College Science Award**, Research Corporation, amount $\$27,500$, 9/2001-2004.
• project title: *Effect of geometric frustration and disorder on weakly coupled spin chains.*

Research Experience

7/12 - present	*Professor*, University of Utah
8/04 - 6/12	*Associate Professor*, University of Utah
9/00 - 8/04	*Assistant Professor*, Hofstra University, NY
9/98 - 8/00	*Postdoctoral Associate*, Yale University, New Haven, CT
11/96 - 8/98	*Postdoctoral Fellow*, University of Florida, Gainesville, FL
9/95 - 10/96	*Postdoctoral Fellow*, University of California, Davis, CA
4/93 - 8/95 5/92 - 10/92	*Postdoctoral Research Associate*, Texas Center for Superconductivity and University of Houston, Houston, TX
9/91 - 5/92	*Junior Researcher*, Institute for High Pressure Physics, Moscow
6/88 - 8/91	*Graduate Student*, Moscow Institute of Physics and Technology, Moscow, Russia

Teaching Experience

Spring 2026	Physics 7120: *Electrodynamics II*
Fall 2025	Physics 7510: *Advanced Solid State I*
Spring 2024	Physics 7120: *Electrodynamics II*
Spring 2024	Physics 7510: *Advanced Solid State I*
Fall 2022	Physics 7640: *Quantum Field Theory I*
Fall 2022	Physics 7510: *Advanced Solid State I*
Spring 2022	Physics 7310: *Statistical Mechanics*
Fall 2021	Physics 7650: *Quantum Field Theory II*
Spring 2021	Physics 7310: *Statistical Mechanics*
Spring 2021	Physics 7640: *Quantum Field Theory I*,
Fall 2020	`no teaching due to the covid-caused schedule disruption`
Spring 2020	Physics 7310: *Statistical Mechanics*
Fall 2019	Physics 7510: *Advanced Solid State I: Physics of Modern Materials*
Spring 2019	Physics7120: *Electrodynamics II*
Fall 2018	Physics7110: *Electrodynamics I*
Spring 2018	Physics7120: *Electrodynamics II*
Fall 2017	`no teaching, sabbatical semester`
Spring 2017	Physics7120: *Electrodynamics II*
Fall 2016	Physics5510: *Solid State Physics I*
Spring 2016	Physics7120: *Electrodynamics II* (together with E. Mishchenko)
Fall 2015	Physics7110: *Electrodynamics I* (together with E. Mishchenko)
Spring 2015	Physics7510: *Advanced Solid State I*
Fall 2014	Physics7640: *Quantum Field Theory I*
Spring 2014	Physics7120: *Electrodynamics II* (together with E. Mishchenko)
Fall 2013	Physics7110: *Electrodynamics I* (together with E. Mishchenko)
Spring 2013	Physics7910: *Quantum Magnetism*
Fall 2012	Physics7510: *Advanced Solid State: Physics of Modern Materials*
Spring 2012	Physics5020: *Theoretical E&M and Statistical Mechanics*
Fall 2011	Physics7110: *Electrodynamics I*
Spring 2011	Physics5020: *Theoretical E&M and Statistical Mechanics*
Fall 2010	`no teaching, sabbatical semester`
Spring 2010	Physics5460: *Quantum and Statistical Mechanics*
Fall 2009	Physics2110: *General Physics with Calculus I*
Spring 2009	Physics5460: *Quantum and Statistical Mechanics*
Fall 2008	Physics7910: *Quantum Magnetism* (special topics course)
Spring 2008	Physics5460: *Quantum and Statistical Mechanics*
Fall 2007	Physics5450: *Quantum Mechanics*
Spring 2007	Physics5460: *Quantum and Statistical Mechanics*
Fall 2006	Physics5450: *Quantum Mechanics*
Spring 2006	Physics5460: *Quantum and Statistical Mechanics*
Fall 2005	Physics5450: *Quantum Mechanics*
Spring 2005	Physics1500: *Preparation for College Physics*
9/00 - 8/04	Lecturing undergraduate physics, Hofstra University: *Mechanics, Electricity and Magnetism, Physics Laboratory.*
2/99-5/99	Lecturer for graduate *Solid State Physics*, Yale University.

Students	8/2024 - present	Xinyuan Xu, Ph.D. graduate student
	8/2018 - 8/2022	Ren-Bo Wang, Ph.D. graduate student (currently postdoc in KITP)
	Fall2019 - Spring2020	Megha Agarwal, research project (currently postdoc at Stockholm Univ)
	6/2014 - 6/2019	Hassan Allami, Ph.D. grad student (currently postdoc at Univ of Ottawa)
	Spring2017-Spring2018	Zhu-Xi Luo, research project (currently postdoc at KITP, UCSB)
	Spring2017	Ethan Lake, research project (currently graduate student at MIT)
	5/2015 -7/2015	Ethan Lake, undergrad REU student
	9/1/2013 - 9/30/2017	Wen Jin, Ph.D. graduate student (currently postdoc at Univ of Waterloo)
	3/1/2014 - 7/1/2017	Ran Tao, M.S. graduate student
	1/1/2011 - 6/2012	Rachel Glenn, Ph.D. graduate student, research project
	8/1/2009 - 7/31/2010	Shane Head, M.S. graduate student
	6/1 - 8/1/2009	Shane Head, undergraduate student
	6/1/2005-7/31/2008	Jianmin Sun, Ph.D. graduate student (Machine Learning engineer)
Postdocs	7/1/2005-9/1/2008	Dr. Suhas Gangadharaiah, postdoc (now faculty at IISER Bhopal, India)

Department Service	2005-2006	*Committees*: Common Exam, Library (chair), Policy Board, Research Council, Solid State Seminar, Theoretical High Energy and Astrophysics (HEAP) faculty Search.
	2006-2007	*Committees*: Advising (graduate), Common Exam, Library (Chair), Solid State Seminar.
	2007-2008	*Committees*: Admissions, Advising (graduate), Common Exam, Policy Board.
	2008-2009	*Committees*: Admissions, Common Exam (chair), Library (chair), Solid State Seminar (chair).
	2006-2008	*College of Science:* Intellectual Explorations Area Committee.
	2010-2011	*Committees*: Admissions (chair), Common Exam, Futures.
	2011-2012	*Committees*: CMT search (chair), Futures.
	2013-2014	*Committees*: Common Exam, Futures, CM seminar, Program Development (grad), Web Page, Policy Board.
	2014-2015	*Committees*: Awards, CM seminar, Web Page, Policy Board.
	2015-2016	*Committees*: condensed matter seminar, Policy board.
	2017-2018	*Committees*: condensed matter seminar.
	2018-2019	*Committees*: CMT search committee (chair).
	2019-2022	*Committees*: Futures committee.
	2021-2022	*Committees*: High Energy / Condensed Matter Theory search, Futures
	2023-2024	*Committees*: Condensed Matter Experiment search; Awards (students); Futures; Graduate Program, Curriculum, and Recruitment.
	2024-2025	*Committees*: Graduate Program, Curriculum, and Recruitment; Utah Physics Circle; Graduate Admissions; University Academic Senate
	2025-2026	*Committees*: Outreach (Utah Physics Circle), chair; Graduate Admissions; University Academic Senate.

Community Service	2024	Co-organizer of the Pollica Physics Center workshop *Exotic quantum matter from quantum spin liquids to novel field theories* `https://agenda.infn.it/event/39804/`; Pollica, June 17 - 28, 2024, Italy;
	2024-2025	Focus Topic co-organizer, GMAG unit, APS March Meeting 2025;
	2022/2023	*Co-organizer* of the KITP program *New Spin on Quantum Magnetism* `https://www.kitp.ucsb.edu/activities/qmagnets23`; Kavli Institute for Theoretical Physics, UCSB; Santa Barbara, July 17, 2023 - September 8, 2023;
	2023 - present	Utah Physics Circle `https://www.physics.utah.edu/utah-physics-circle/`, a physics problem-solving club for high school students, *co-organizer*;
	Fall 2015	Focus Topic organizer for the March 2016 APS meeting, topic "10.1.6 Frustrated Magnetism" (GMAG/DMP);
	01/2003 - 12/2005	APS Committee for the International Freedom of Scientists (CIFS);
	10/97 - present	Referee for *Physical Review Letters* and *Physical Review B*, *Europhysics Letters* and *Journal of Physics: Condensed Matter*; *New Journal of Physics*, *Nature*, *Nature Physics*, *SciPost*, *Annals of Physics*;
	1986 - 1990	Physics and Mathematics Instructor, one month per year, Summer School for talented high-school students, Krasnoyarsk, Russia.

Funding	Dec. 2025 (pending)	∗ Pre-proposal to Schwinger Foundation, `https://schwingerfoundation.org/`, • title: *Quantum electrodynamics of frustrated spin models*, amount $200,000.
	10/01/2025 - 09/30/2027	∗ Co-PI, with Prof. A. Keselman (Technion, Israel), United States-Israel Binational Science Foundation, **BSF** award # 2024200, amount $90,000: • title: *Quantum phases and excitations of magnetized frustrated magnets.*
	12/01/2019-11/30/2023	∗**NSF DMR**, award # 1928919, amount $330,000: • title: *Collective Modes and Electrodynamics of Interacting Spin Liquids.*
	11/03-22/2017	∗**Japan Society for the Promotion of Science** Invitational award: Short term visit – travel expences and daily allowance for 19 days.
	3/2016-2/2019	∗**NSF DMR** award # 1507054, amount $306,000: • title: *Order-by-frustration: emergent condensed states of frustrated magnets.*
	9/2012-8/2016	∗**NSF DMR** award # 1206774, amount $270,000: • project title: *Frustrated magnetism in spin-orbit coupled materials.*
	2/2014-2015	∗ Co-PI, with Prof. D. Pesin, of Utah MRSEC Seed Grant, amount $24,000: • *Transport and correlated phases in system with strong spin-orbit interaction*
	5/2009-2013	∗**NSF DMR**, award # 0808842, amount $210,000: • title: *Competing orders in frustrated magnets and nanostructures,* supports graduate and summer undergraduate students.
	9/2005-8/2007	∗**The Petroleum Research Fund**, American Chemical Society, amount $80,000. • title: *Frustrated magnetism as a problem of coupled spin chains.*
	9/2001-2004	∗**Cottrell College Science Award**, Research Corporation, amount $27,500. • title: *Effect of geometric frustration and disorder on weakly coupled spin chains.*
	2001-2003	∗ **ITP Scholar**, Kavli Institute for Theoretical Physics, UCSB.

Visiting Positions	07/2004	Visiting Scientist, Condensed Matter Group, International Centre for Theoretical Physics (ICTP), Trieste, Italy
	05/12/ - 06/12/2005	Visiting Scientist, Theory Institute, Brookhaven National Laboratory, Upton, NY
	03/10-22/2008	Visiting Scientist, Condensed Matter Theory Laboratory, RIKEN, Japan (two weeks including Spring Break).
	07/14-26/2008	Visiting Scientist, Joint Theory Institute, University of Chicago and Argonne National Laboratory.
	07/10/2010 - 10/06/2010	Visiting Professor, Condensed Matter Group, Max Planck Institute for Physics of Complex Systems, Dresden, Germany.
	05/13 - 05/27/2012	Participant in program "New quantum states of matter in/out of equilibrium", Galileo Galilei Institute for Theoretical Physics, Florence, Italy
	06/05 - 07/05/2017	Visiting professor, Laboratoire de Physique Théorique et Modélisation (LPTM), Université de Cergy-Pontoise, Cergy, Ile-de-France, France.
	11/02-22/2017	Visiting Scientist, Condensed Matter Theory Laboratory, RIKEN, Japan.

SEMINARS and INVITED TALKS

1. *$J_1 - J_2$ triangular lattice antiferromagnet in magnetic field*, invited talk at the international workshop "Current trends in strongly correlated and frustrated systems", Max Planck Institute for the Physics of Complex Systems, Dresden, Germany, November 10-14, 2025.

2. *Magnetized quantum magnets: spinons, their bound states, and quantum chiral solitons*, seminar at the Department of Physics and Astronomy, Purdue University, October 31, 2025.

3. *Spinon spin current*, seminar at the Department of Physics and Astronomy, University of Utah, November 12, 2024.

4. *The Magnanimous Spinon*, blackboard talk at KITP's Locals' Lunch, September 16, 2024.

5. *Magnetized quantum spin liquids*, seminar at the Department of Physics and Astronomy, Johns Hopkins University, July 29, 2024.

6. *Dynamical response of magnetized spin liquids*, invited talk at the ICTP Conference on "Fractionalization and Emergent Gauge Fields in Quantum Matter", ICTP, Trieste, Italy, December 7, 2023.

7. *The Magnanimous Magnon*, KITP Blackboard Lunch talk, KITP, September 5, 2023.

8. *QMagnets23 organizational meeting*, KITP Program "A New Spin on Quantum Magnets" (Jul 17 - Sep 8, 2023; `https://www.kitp.ucsb.edu/activities/qmagnets23`), KITP, July 18, 2023.

9. *The modern revival of electron spin resonance in quantum magnets*, online talk at the JITCP seminar, Department of Physics, The University of Hong Kong, May 17/18, 2023.

10. *Spectroscopic and transport signatures of one-dimensional spinon liquid*, condensed matter seminar at Department of Physics and Astronomy, UC Irvine, February 8, 2023.

11. *Electrical resistivity of the Kondo-like triangular lattice antiferromagnet in a magnetic field – the story of EuC_6*, condensed matter seminar, University of Utah, September 6, 2022.

12. *Hydrodynamic theory of interacting spinon liquid in quantum spin chains*, invited talk at the ICTP conference "Strongly Correlated Matter: from Quantum Criticality to Flat Bands", ICTP, Trieste, Italy, August 25, 2022; `https://indico.ictp.it/event/9822/other-view?view=ictptimetable`.

13. *Electron spin resonance of interacting spinons*, invited talk at the "Topology, Geometry, and Physics" symposium celebrating Prof. Yong-Shi Wu's 80th Birthday [online], August 4, 2022; `https://sites.google.com/view/yswu2022`.

14. *Hydrodynamic theory of interacting spinon liquid and its predictions for experiments on quantum spin chains*, talk at the conference Highly Frustrated Magnetism (HFM 2022), Sorbonne University, Paris, France, June 19-25, 2022; `https://hfm22.physics.cnrs.fr/`.

15. *Electrical resistivity of the Kondo-like triangular lattice antiferromagnet in a magnetic field: the story of* EuC_6, condensed matter seminar, School of Physics and Astronomy, University of Minnesota, May 4, 2022.

16. *Electrical resistivity of the Kondo-like triangular lattice antiferromagnet in a magnetic field: the story of* EuC_6, invited talk at the International workshop "Trends in Quantum Magnetism", Congresso Stefano Franscini, Ascona, Switzerland, April 24-27, 2022; `https://tqm2020.ethz.ch/`.

17. *Electron spin resonance spectroscopy of quantum spin liquids*, colloquium at the Department of Physics, University of Virginia, April 8, 2022; `https://www.phys.virginia.edu/Announcements/talk-list.asp?SELECT=SID:4131`

18. *Electron spin resonance of the interacting spinon liquid*, condensed matter seminar, University of Utah, September 28, 2021.

19. *Collective modes of magnetized spin liquids*, invited talk at the virtual APS March Meeting, March 15-19, 2021.

20. *Collective modes of magnetized spin liquids*, online "Quantum Matter" seminar, Northeastern University, February 17, 2021; YouTube video recording `https://www.youtube.com/watch?v=dRw0LbTVraM`

21. *Collective modes of magnetized spin liquids*, invited talk at KITP program "Correlated Systems with Multicomponent Local Hilbert Spaces", Kavli Institute for Theoretical Physics, UCSB, Santa Barbara, October 29, 2020; `https://online.kitp.ucsb.edu/online/correlated20/starykh/`.

22. *Collective modes of a U(1) magnetized spin liquid*, invited talk at the (cancelled due to covid-19) APS March Meeting, Denver, Colorado, March 2, 2020.

23. *Spintronics meets Quantum Spin Liquids*, informal discussion/talk at KITP program "Spin and Heat Transport in Quantum and Topological Materials", Kavli Institute for Theoretical Physics, UCSB, Santa Barbara, December 13, 2019.

24. *Collective modes of a U(1) magnetized spin liquid*, talk at KITP program "Topological Quantum Matter: Concepts and Realizations", Kavli Institute for Theoretical Physics, UCSB, Santa Barbara, October 17, 2019.

25. *Majorana zero modes at the edge of an interacting quantum wire with spin-orbit interaction*, invited talk at the conference "Spins in a quantum 1D multi-particle environment: from exotic phases and non-trivial topology to protected transport", Ludwig Maximilian University, Munich, Germany, September 2-5, 2019.

26. *Collective modes of magnetized spin liquids*, theory talk at the Aspen Center for Physics, "Quantum Spin Liquids" program, Aspen, Colorado, July 11, 2019.

27. *Quantum spin liquids*, presentation for REU students, University of Utah, June 26, 2019.

28. *Collective modes of magnetized spin liquids*, invited talk at the workshop "Low-dimensional emergent phenomena in correlated systems and topological quantum matter", Tbilisi, Republic of Georgia, June 4, 2019.

29. *Collective modes of magnetized spin liquids*, condensed matter seminar, Department of Physics, University of Minnesota, March 13, 2019.

30. *Spinon magnetic resonance of two-dimensional U(1) spin liquids with Fermi surface*, talk at Highly Frustrated Magnetism (HFM-2018) workshop, UC Davis, July 14, 2018.

31. *Spinon magnetic resonance*, invited talk at TOPMAT workshop, Saclay, France, June 11, 2018.

32. *Spinon magnetic resonance*, invited talk at BigMag@UCSB workshop, UCSB, Santa Barbara, May 17-19, 2018.

33. *Phases of spin chains with uniform Dzyaloshinskii-Moriya interactions*, Condensed Matter Theory Group seminar, RIKEN, Wako, Japan, November 21, 2017.

34. *Chiral liquid phase of simple quantum antiferromagnets*, condensed matter seminar, ISSP, University of Tokyo, Kashiwa campus, Japan, November 20, 2017.

35. *Chiral liquid phase of simple quantum antiferromagnets*, YITP workshop "Novel quantum states in condensed matter", Yukawa Institute for Theoretical Physics, Kyoto University, Japan, November 9, 2017.

36. *Two-magnon condensation route to the chiral spin liquid*, invited talk at the Gordon Godfrey workshop, University of New South Wales, Sydney, Australia, October 31, 2017.

37. *Chiral liquid phase of simple XXZ quantum magnets*, talk at KITP Program "Intertwined Order and Fluctuations in Quantum Materials", Kavli Institute for Theoretical Physics, UCSB, Santa Barbara, October 4, 2017.

38. *Emergent Ising orders of frustrated magnets*, Jussieu, Université Pierre et Marie Curie (Sorbonne University), Laboratoire de Physique Théorique de la Matiere Condensee, Paris, July 3, 2017.

39. *Emergent Ising orders of frustrated magnets*, Laboratoire de Physique Théorique et Modélisation (LPTM), Université de Cergy-Pontoise, Cergy, June 22, 2017.

40. *Emergent Ising orders of frustrated magnets*, invited talk at conference "SPIN COHERENCE, CONDENSATION, AND SUPERFLUIDITY", University of California Gump Station, Moorea, French Polynesia, February 13-17, 2017.

41. *A panoply of orders near quantum Lifshitz point of a frustrated ferromagnet*, tutorial talk at the SPICE workshop "Quantum Spintronics: Spin Transport Through Quantum Magnetic Materials", Johannes Gutenberg University, Mainz, Germany, September 21-23, 2016.

42. *Frustration-driven multi-magnon condensates*, solid state seminar, Department of Physics, ETH, Zurich, Switzerland, September 19, 2016.

43. *Frustration-driven multi-magnon condensates*, invited talk at the workshop "Recent progress in low-dimensional quantum magnetism" (LDQM2016), EPFL, Lausanne, Switzerland, September 5-16, 2016.

44. *Rashba vs Kohn-Luttinger: evolution of p-wave superconductivity in magnetized two-dimensional Fermi gas subject to spin-orbit interaction*, invited talk at the workshop "Multi-component and strongly-correlated superconductors", Nordita, Stockholm, Sweden, July 15, 2016.

45. *Frustration-driven multi-magnon condensates*, seminar at the University of Utah, January 12 (part 1) and January 26 (part 2), 2016.

46. *Frustration-driven multi-magnon condensates and their excitations*, colloquium at Los Alamos National Laboratory, November 18, 2015.

47. *Unusual phases of antiferromagnetic spin chains with uniform Dzyaloshinskii-Moriya interaction*, invited talk a the SPICE workshop "Magnetic Adatoms as Building Blocks for Quantum Magnetism", Mainz, Germany, August 17-20, 2015.

48. *Magnon collapse near the Lifshitz point and multipolar phases of frustrated magnets*, talk at KITP program "New Phases and Emergent Phenomena in Correlated Materials with Strong Spin-Orbit Coupling", Kavli Institute for Theoretical Physics, UCSB, Santa Barbara, July 24, 2015.

49. *Frustration-driven multi magnon condensates and their excitations*, seminar at the Condensed Matter Theory Laboratory, RIKEN, Wako, Japan, March 16, 2015.

50. *Quasi-one-dimensional spin nematic states and their excitations*, invited talk at the APS March Meeting, San Antonio, March 2 - 5, 2015.

51. *Frustration-driven multi magnon condensates and their excitations*, invited talk at the ICTP workshop on Current trends in frustrated magnetism, Jawaharlal Nehru University, New Delhi, India, February 9 - 13, 2015.

52. *Two-magnon instabilities and other surprises in magnetized quantum antiferromagnets*, invited talk at the Conference on Field Theory Methods in Low-Dimensional Strongly Correlated Quantum Systems, ICTP, Trieste, Italy, August 25-29, 2014.

53. *Unusual ordered phases of magnetized frustrated antiferromagnets*, talk at the International Conference on Highly Frustrated Magnetism 2014, University of Cambridge, Queen's College, Cambridge, UK, July 7 - 11, 2014.

54. *Unusual ordered phases of magnetized frustrated antiferromagnets*, invited talk at the workshop "Latest Developments in Frustrated Magnetism:spin liquids, spin glass, spin ice states", University of Warwick, UK, July 5, 2014.

55. *Unusual ordered phases of magnetized frustrated antiferromagnets*, invited talk at 2nd International Symposium on "Novel states in correlated condensed matter - from model systems to real materials", Königstein near Frankfurt am Main, Germany, April 8-10, 2014.

56. *Unusual ordered phases of magnetized frustrated antiferromagnets*, colloquium at the Department of Physics, University of Regensburg, Regensburg, Germany, April 7, 2014.

57. *Unusual ordered phases of magnetized frustrated antiferromagnets*, condensed matter seminar, Department of Physics, University of Waterloo, Waterloo, Canada, March 20, 2014.

58. *Frustrated magnetism via bold diagrammatic Monte Carlo*, invited talk at SIGN 2014, an international EMMI Workshop on the "Sign Problem in QCD and beyond", GSI Helmholtzzentrum für Schwerionenforschung GmbH, Darmstadt, Germany, 18-21 February 2014.

59. *Spin-current and other unusual phases in magnetized triangular lattice antiferromagnets*, talk at the International Seminar and Workshop on "Spin Orbit Entanglement: Exotic States of Quantum Matter in Electronic Systems", Max Planck Institute for Physics of Complex Systems, Dresden, Germany; July 15 - 27, 2013.

60. *Surprises at the magnetization plateau*, condensed matter seminar, Department of Physics and Astronomy and Institute for Quantum Matter, Johns Hopkins University, Baltimore, MD; April 29, 2013.

61. *Surprises at the magnetization plateau*, condensed matter seminar, Department of Physics, Texas A& M University, College Station, TX; March 29, 2013.

62. *Spin ice and magnetic monopoles*, undergraduate seminar, University of Utah, November 15, 2012.

63. *Magnetization Plateaux of Frustrated Antiferromagnets*, invited plenary talk at the APS Four Corners Section regional meeting, New Mexico Tech, Socorro, NM, October 26-27, 2012.

64. *Electron spin resonance of spinon gas*, invited talk at the KITP Conference "Exotic Phases of Frustrated Magnets", KITP, UCSB, Santa Barbara, October 8-12, 2012.

65. *Exotic ordered phases of triangular magnets*, invited talk at the Quantum Condensed Matter Division and the Virtual Institute Joint Meeting "New States of Matter and their Excitations", Oak Ridge National Laboratory, Knoxville, TN, September 12-14, 2012.

66. *Fate of 1/3 magnetization plateau in spatially anisotropic triangular lattice antiferromagnets*, talk at the Highly Frustrated Magnetism conference (HFM 2012), McMaster University, Hamilton, ON, Canada, June 4 - 8, 2012.

67. *Magnetization plateau and other unusual phases of a spatially anisotropic quantum antiferromagnet on triangular lattice*, invited talk at the conference on "New quantum states of matter in and out of equilibrium", the Galileo Galilei Institute for Theoretical Physics, Florence, Italy, May 25, 2012.

68. *Breaking the spin waves: spinons and other strange excitations of frustrated quantum antiferromagnets*, colloquium at the Department of Physics, Brigham Young University, Provo, March 28, 2012.

69. *Breaking the spin waves: spinons in Cs_2CuCl_4 and elsewhere*, colloquium at the Department of Physics, Iowa State University, Ames, February 13, 2012.

70. *Triangular lattice antiferromagnet in magnetic field: ground states and spin excitations*, seminar at P.L.Kapitza Institute for Physical Problems, Moscow, Russia, October 4, 2011.

71. *Breaking the spin waves: spinons in Cs_2CuCl_4 and elsewhere*, invited talk, International Symposium Spin Waves 2011, St. Petersburg, Russia, June 5-11, 2011.

72. *Modes of magnetic resonance in spin liquids with spinon excitations*, invited talk, International Conference on Novel Phenomena in Frustrated Systems, Santa Fe, New Mexico, May 23-27, 2011.

73. *Modes of magnetic resonance in spin liquids with spinon excitations*, condensed matter seminar, Department of Physics, University of Arizona, Tuscon, April 21, 2011.

74. *Magnetization plateaux in triangular lattice antiferomagnets*, condensed matter seminar, Department of Physics, University of Waterloo, Canada, December 14, 2010.

75. *Magnetization plateaux in triangular lattice antiferomagnets*, condensed matter seminar, Department of Physics, University of Toronto, Canada, December 13, 2010.

76. *Triangular lattice antiferromagnets – open questions*, Workshop on "Disentangling quantum many-body systems", KITP, Santa Barbara, November 11, 2010.

77. *Spin-orbit-induced spin-density wave in quantum wires and spin chains*, seminar, Dahlem Centex for Complex Quantum Systems, Freie Universitat Berlin, Berlin, Germany, September 29, 2010.

78. *Magnetization plateaux in triangular lattice antiferromagnets*, seminar, Institute for Complex Magnetic Materials, Helmholtz-Zentrum-Berlin, Berlin, Germany, September 21, 2010.

79. *Spatially anisotropic triangular antiferromagnet in a magnetic field*, invited talk, workshop on "Emergent Quantum States in Complex Correlated Matter", Max Planck Institute for Physics of Complex Systems, Dresden, Germany; August 23 - 27, 2010.

80. *Quantum (and classical) physics of* Cs_2CuCl_4, seminar, Leibniz Institute for Solid State and Materials Research, IFW Dresden, Germany; August 17, 2010.

81. *Quasi-one-dimensional version of quantum kagomé antiferromagnet*, seminar, Max Planck Institute for Physics of Complex Systems, Dresden, Germany; August 16, 2010.

82. *Quantum (and classical) physics of* Cs_2CuCl_4, seminar, High Magnetic Field Laboratory, Forschungszentrum Dresden - Rossendorf, Germany; July 21, 2010.

83. *Frustrated quasi-one-dimensional quantum magnet in magnetic field*, invited talk, workshop on "Correlated phenomena in low-dimensional systems", Max Planck Institute for Physics of Complex Systems, Dresden, Germany; July 15, 2010.

84. *Ground States and Excitations of Spatially Anisotropic Quantum Antiferromagnets*, invited talk, "Workshop on the Heisenberg model: past, present and future", International Center on Condensed Matter Physics, University of Brasilia, Brasilia, Brasil; July 20 - 27, 2009.

85. *Spatially anisotropic spin-1/2 Heisenberg kagomé antiferromagnet*, invited talk, APS March Meeting, Pittsburgh, PA; March 19, 2009.

86. *Ice, Spin and Spin Ice*, undergraduate seminar, University of Utah, December 4, 2008.

87. *Spinons and triplons in a spatially anisotropic triangular antiferromagnet*, condensed matter seminar, Department of Physics, University of Wisconsin, Madison, October 23, 2008.

88. *Spinons and triplons in a spatially anisotropic triangular antiferromagnet*, condensed matter seminar, Department of Physics, École Polytechnique Fédérale de Lausanne, Lausanne, Switzerland, July 1, 2008.

89. *Spinons and triplons in a spatially anisotropic triangular antiferromagnet*, Max-Planck-Institut für Physik komplexer Systeme, Dresden, Germany, June 18, 2008.

90. *Spin-orbit induced spin-density wave in an interacting quantum wire*, Institute for Solid State Physics, University of Tokyo, Japan, March 18, 2008.

91. *Spinons and triplons in a spatially anisotropic triangular antiferromagnet*, Condensed Matter Theory Laboratory, RIKEN, Japan, March 17, 2008.

92. *Spinons and triplons in a spatially anisotropic triangular antiferromagnet*, solid state seminar, Department of Physics, Texas A & M University, November 27, 2007.

93. *Spinons and triplons in a spatially anisotropic triangular antiferromagnet*, solid state seminar, Department of Physics, Stony Brook University, November 9, 2007.

94. *Spin-orbital effects in interacting electrons: quantum dots and wires*, talk at the "Hard Times" seminar, Department of Physics, UCSB, October 19, 2007.

95. *Orders and Excitations in Frustrated Quasi-One-Dimensional Antiferromagnets*, talk at the "Moments and Multiplets in Mott Materials" Program, Kavli Institute for Theoretical Physics, University of California at Santa Barbara, October 18, 2007.

96. *Spontaneous Dimer Order in Spatially Anisotropic Antiferromagnets*, talk at the "Strongly Correlated Phases in Condensed Matter and Degenerate Atomic Systems" Program, Kavli Institute for Theoretical Physics, University of California at Santa Barbara, May 21, 2007. (the talk is available online: http://online.kitp.ucsb.edu/online/coldatoms07/starykh/)

97. *Heisenberg antiferromagnet on an anisotropic triangular lattice: spin order and excitations*, seminar, Argonne National Laboratory, March 23, 2007.

98. *Spin-1/2 Heisenberg antiferromagnet on an anisotropic triangular lattice*, invited talk, APS March Meeting, Denver, Colorado, March 5, 2007.

99. *Flat spin wave dispersion in a triangular antiferromagnet*, regular talk, APS March Meeting, Denver, Colorado, March 5, 2007.

100. *Ice, spin and spin ice*, undergraduate seminar, University of Utah, September 21, 2006.

101. *Magnons, rotonsm and spinons in quantum antiferromagnets*, colloquium, University of Utah, September 14, 2006.

102. *Spin density wave in a quantum wire with spin-orbit interaction*, seminar, UC Santa Barbara, June 9, 2006.

103. *Spin density wave in a quantum wire with spin-orbit interaction*, seminar, UC Irvine, June 7, 2006.

104. *Spin density wave in a quantum wire with spin-orbit interaction*, condensed matter seminar, University of Utah, April 25, 2006.

105. *A spin on ice*, undergraduate seminar, University of Utah, October 27, 2005.

106. *Spontaneous dimerization in frustrated magnets*, seminar, University of Colorado, Boulder, October 6, 2005.

107. *Dimerized phases of quasi-one-dimensional magnets*, seminar at the Brookhaven National Laboratory, June 9, 2005.

108. *Spin fluctuations in one dimension: NMR in spin chains and tunneling in quantum wires*, condensed matter seminar, University of Utah, April 5, 2005.

109. *One-dimensional view of frustrated magnets*, invited talk at the International Workshop on Physics of Strongly Correlated Electron Systems, Yukawa Institute for Theoretical Physics, Kyoto University, Kyoto, Japan, November 1-19, 2004.

110. *Quantum mechanics of electrical current circuits*, undergraduate seminar, October 21, 2004.

111. *One-dimensional view of frustrated magnets*, invited talk at the International Workshop on Evolution of Quantum Effects from the Nano- to the Macroscale, Cargese, Corsica, France, May 17-21, 2004.

112. *One-dimensional view of frustrated magnets*, seminar, Yale University, Aprill 8, 2004.

113. *Breaking the spin waves*, colloquium, Department of Physics, University of Utah, March 8, 2004.

114. *Zeeman splitting of zero-bias anomaly in Luttinger liquids*, invited talk at the International Workshop on Field Theory Methods in Correlated Nanoscale Systems, Brookhaven National Laboratory, August 26-30, 2003.

115. *A 'one-dimensional' view of geometrically frustrated magnets*, seminar, The Johns Hopkins University, October 2, 2002; and SUNY Stony Brook, May 3, 2002.

116. *ibid*, invited talk at the Workshop on Quantum Spins, Lorentz Center, Leiden University, August 5-16, 2002.

117. *A 'one-dimensional' approach to frustrated magnets*, seminar, University of Massachussets, Amherst, March 2002.

118. *Spinons in a crossed-chains model of a 2D spin liquid*, seminar, Leiden University, The Netherlands, July 2001.

119. *ibid*, Utrecht University, The Netherlands, July 2001.

120. *Thermally fluctuating superconductors in d=2*, seminar, Utrecht University, The Netherlands, June 2000.

121. *ibid*, University of Groningen, The Netherlands, June 2000.

122. *ibid*, Leiden University, The Netherlands, July 2000.

123. *Electrons on the line: field theory meets experiments*, Colloqium, Utrecht University, the Netherlands, June 2000.

124. *Electrons on the line: spin chains, quantum wires and beyond*, Colloqium, University of California at Riverside, February 2000.

125. *ibid*, colloqium at Clark University, Worchester, MA, February 2000.

126. *Logarithmic corrections to quantum-critical scaling in the S=1/2 Heisenberg chain*, Talk at the ITP Conference on Quantum Magnetism, University of California at Santa Barbara, August 16-20, 1999.

127. *Luttinger liquid behavior of quantum wire*, seminar, Argonne National Laboratory, July, 1999.

128. *Finite temperature dynamics of the spin-1/2 Heisenberg chain*, Invited Talk at the March Meeting of the American Physical Society, Los Angeles, CA, March 16, 1998.

129. *Quantum Phase Transitions and Quantum Antiferromagnets*, Colloqium at the Department of Physics, University of Colorado, Boulder, March 4, 1998.

130. *Conductance of the Mott quantum wire*, seminar, Department of Physics, University of Illinois at Urbana-Champaign, December 10, 1997.

131. *ibid*, Department of Physics, University of Chicago, December 9, 1997.

132. *ibid*, Department of Physics, University of Indiana at Bloomington, December 5, 1997.

133. *ibid*, Department of Physics, University of Wisconsin at Madison, December 4, 1997.

134. *Origin of spin gap behavior of* CaV_4O_9, seminar, Texas Center for Superconductivity, University of Houston, July 23, 1996.

135. *Unusual properties of low-dimensional quantum antiferromagnets*, seminar at the Department of Physics, Yale University, April 24, 1996.

136. *Spin gap behavior of two-dimensional quantum antiferromagnet and effect of spin-lattice coupling*, University of Chicago, April 22, 1996.

137. *Quantum Antiferromagnet and Confinement of Spinons in the* CP^{N-1} *Model*, Department of Physics, University of Pittsburgh, Pittsburgh, PA, 1994.

PUBLICATIONS of Oleg A. Starykh

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2. Leandro M. Chinellato, Oleg A. Starykh, Cristian D. Batista, *Dynamics of Quantum Chiral Solitons*, https://www.arxiv.org/abs/2512.08220 (2025).

3. Anna Keselman, Xinyuan Xu, Hao Zhang, Cristian D. Batista, Oleg A. Starykh, *J_1-J_2 Triangular Lattice Antiferromagnet in a Magnetic Field*, https://www.arxiv.org/abs/2512.02150 (2025).

4. Chuang Chen, Urban F. P. Seifert, Kexin Feng, Oleg A. Starykh, Leon Balents, Zi Yang Meng, *Emergent gauge flux in QED_3 with flavor chemical potential: application to magnetized U(1) Dirac spin liquids*, https://arxiv.org/abs/2508.08528 (2025).

5. P. Mokhtari, S. Galeski, U. Stockert, S. E. Nikitin, R. Wawrzynczak, R. Kuechler, M. Brando, L. Vasylechko, O. A. Starykh, E. Hassinger, *1/5 and 1/3 magnetization plateaux in the spin-1/2 chain system $YbAlO_3$*, arXiv:2412.21144, Phys. Rev. Lett. **135**, 076704 (2025), https://journals.aps.org/prl/abstract/10.1103/grfl-37g2; https://www.arxiv.org/abs/2412.21144 (2024).

6. Ren-Bo Wang, Naveen Nishad, Anna Keselman, Leon Balents, Oleg A. Starykh, *Spin Seebeck effect of interacting spinons*, Phys. Rev. B **112**, L060402 (2025), https://journals.aps.org/prb/abstract/10.1103/ncdj-9f2t; (aka *Spinon spin current*, arXiv:2409.08327 , https://arxiv.org/abs/2409.08327 (2024)).

7. Oleg Starykh, *Viewing a Quantum Spin Liquid through QED*, Physics **17**, 63 (2024); https://physics.aps.org/articles/v17/63.

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9. Ren-Bo Wang, Anna Keselman, Oleg A. Starykh, *Hydrodynamics of interacting spinons in the magnetized spin-1/2 chain with the uniform Dzyaloshinskii-Moriya interaction*, Phys. Rev. B **105**, 184429 (2022), https://journals.aps.org/prb/abstract/10.1103/PhysRevB.105.184429; arXiv:2201.10570 (2022).

10. A. L. Chernyshev and O. A. Starykh, *Roller-Coaster in a Flatland: Magnetoresistivity in Eu-intercalated Graphite*, Phys. Rev. X **12**, 021010 (2022), https://journals.aps.org/prx/abstract/10.1103/PhysRevX.12.021010; arXiv:2109.10916 (2021).

11. Kirill Yu. Povarov, Timofei A. Soldatov, Ren-Bo Wang, Andrey Zheludev, Alexander I. Smirnov, Oleg A. Starykh, *Electron Spin Resonance of the Interacting Spinon Liquid*, Phys. Rev. Lett. **128**, 187202 (2022) https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.128.187202. [See also Synopsis accompanying the paper in Physics magazine https://physics.aps.org/articles/v15/s59], arXiv:2108.02835 (2021).

12. Chengkang Zhou, Zheng Yan, Kai Sun, Oleg A. Starykh, Zi Yang Meng, *Amplitude Mode in Quantum Magnets via Dimensional Crossover*, Phys. Rev. Lett. **126**, 227201 (2021); arXiv:2007.12715 (2020), https://arxiv.org/abs/2007.12715.

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15. Leon Balents and Oleg A. Starykh, *Collective spinon spin wave in a magnetized U(1) spin liquid*, Phys. Rev. B **101**, 020401(R) (2020); arXiv:1904.02117 (2019), https://arxiv.org/abs/1904.02117.

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21. Wen Jin and Oleg A. Starykh, *Phase diagram of weakly coupled Heisenberg spin chains subject to a uniform Dzyaloshinskii-Moriya interaction*, Phys. Rev. B **95**, 214404 (2017); arxiv:1701.01465 (2017), https://arxiv.org/abs/1701.01465.

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23. Ethan Lake*, Caleb Webb*, D. A. Pesin, O. A. Starykh, *Rashba vs Kohn-Luttinger: evolution of p-wave superconductivity in magnetized two-dimensional Fermi gas subject to spin-orbit interaction*, Phys. Rev. B **93**, 214516 (2016); arXiv:1602.02741 (2016) [* = undergraduate student], https://arxiv.org/abs/1602.02741.

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33. Zhihao Hao and Oleg A. Starykh, *Half-metallic magnetization plateaux*, Phys. Rev. B **87**, 161109(R) (2013); arxiv:1207.7124 (2012), https://arxiv.org/abs/1207.7124.

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41. Oleg A. Starykh, *Spin-density wave in a quantum wire*, - Book Chapter in *Handbook of Nanophysics* vol.**4**: *Nanotubes and Nanowires*, Klaus D. Sattler (Editor), published by CRC Press (September 3, 2010), ISBN-10: 142007542X, page 30-1, `http://www.physics.utah.edu/~starykh/research/papers/Starykh_Ch30.pdf`.

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